

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence

COURSE CODE: CSA17

COURSE OUTCOME COVERED

CO1: Apply AI basics such as intelligent agents and uninformed search methods to solve problems effectively. (BL3)

Assessment Tool Weightage: 10%

QUESTIONS: 2

Duration : 45 Minutes
Total Marks:20

Annexure A

Descriptive Written Test

Code of Conduct:

*I Karanam Bhaskar Gnanesh(192324080) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.*

Karanam Bhaskar Gnanesh
Signature of the student:

Annexure A

Descriptive Written Test

1. A smart home automation system is being designed to manage lighting, temperature, and security based on user behavior and environmental conditions. The system must respond to real-time inputs such as motion detection, temperature changes, and time of day while also learning user preferences over time. In this context, explain the different types of intelligent agents (simple reflex, model-based, goal-based, and utility-based agents) and analyze how each type would function within this system. Justify which type of agent or hybrid approach would be most effective for ensuring comfort, energy efficiency, and security.
2. A robot is placed in an unknown maze and must find a path to the exit without prior knowledge of the layout. Explain how uninformed search strategies like Uniform Cost Search (UCS) and Iterative Deepening Search (IDS) can help solve this problem. Discuss the advantages and disadvantages of these algorithms in terms of time and space complexity. Relate the problem to different types of intelligent agents and explain how agent design affects decision-making. Suggest which combination of agent type and search strategy would be most effective and justify your choice.

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Criteria	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)	Max Marks
Understanding of Concepts	25%	Demonstrates thorough, accurate understanding of concepts	Good understanding with minor gaps	Basic understanding with some misconceptions	Poor understanding; major misconceptions
Explanation	25%	Detailed, well-explained answers with relevant examples	Clear explanation with adequate details	Limited explanation; lacks depth	Poor or unclear explanation
Application	20%	Effectively applies concepts with strong analytical ability	Applies concepts with minor errors	Limited application with weak analysis	No application or incorrect analysis
Organization	15%	Well-structured answers with logical flow and coherence	Organized with minor issues in flow	Basic structure, lacks clarity	Poor organization, incoherent answers
Accuracy	10%	Completely accurate and covers all required points	Mostly accurate with minor omissions	Partially correct with missing points	Incorrect answers with major gaps
Clarity	5%	Clear, neat, professional presentation	Understandable with minor issues	Limited clarity, readability issues	Poorly written, difficult to understand

Question 1

A Smart Home Automation System Using Intelligent Agents

Introduction

A Smart Home Automation System is an intelligent system that automatically controls household devices such as lights, fans, air conditioners, heaters, security cameras, door locks, and other appliances based on user behavior and environmental conditions. The main objective of a smart home is to provide comfort, security, convenience, and energy efficiency while minimizing human intervention.

Artificial Intelligence (AI) plays an important role in smart home automation by enabling the system to learn user preferences, respond to changing conditions, and make intelligent decisions. AI systems use intelligent agents to observe the environment, process information, and perform appropriate actions.

An intelligent agent is a software or hardware entity that perceives its environment through sensors and acts upon that environment using actuators to achieve specific goals.

Components of a Smart Home Automation System

Sensors

Sensors collect information from the environment.

Examples include:

- Motion sensors
- Temperature sensors
- Smoke detectors
- Light sensors
- Door sensors
- Humidity sensors

Controller (Intelligent Agent)

The controller processes the collected data and decides which action should be taken.

Examples:

- Turn lights ON
- Increase AC temperature
- Lock doors
- Trigger alarms

Actuators

Actuators perform physical actions.

Examples:

- Smart lights
- Smart fans
- Smart locks
- Smart curtains
- Smart thermostat

Types of Intelligent Agents

Artificial Intelligence defines four major types of intelligent agents.

1. Simple Reflex Agent

Definition

A Simple Reflex Agent makes decisions based only on the current situation. It does not remember previous events or learn from experience. It follows simple "Condition–Action" rules.

Example:

IF Motion Detected
THEN Turn ON Light

Working in Smart Home

Suppose it is nighttime.

The motion sensor detects movement.

Immediately,

- Lights turn ON
- Corridor lights remain ON for 2 minutes
- Then automatically turn OFF

Another example

IF Temperature > 30°C

Turn ON Air Conditioner

Advantages

- Very fast response
 - Easy to design
 - Low computational cost
 - Suitable for simple automation
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Disadvantages

- Cannot learn user habits
 - Cannot remember previous actions
 - Poor decision-making in complex situations
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2. Model-Based Reflex Agent

Definition

A Model-Based Reflex Agent maintains an internal model (memory) of the environment.

It considers:

- Current state
 - Previous state
 - Environmental changes before making decisions.
-

Working in Smart Home

Suppose

The front door opens.

The agent remembers

- Owner is not at home
- Security mode is enabled
- Instead of simply opening the door, the system
- Turns ON cameras
- Records video

- Sends mobile notification
 - Activates alarm if unknown person enters
- Another example
The room temperature is decreasing.
The agent predicts
"It will become cold soon."
So it switches ON the heater before temperature becomes uncomfortable.
-

Advantages

- Better than simple reflex
 - Uses memory
 - Handles partially observable environments
 - Improves security
-

Disadvantages

- More complex
 - Requires storage
 - Higher computation
-

3. Goal-Based Agent

Definition

A Goal-Based Agent chooses actions that help achieve a specific objective.
It plans before acting.

Example Goal

Maintain room temperature between
22°C and 24°C

Working

Suppose

Current temperature

32°C

Goal

24°C

The agent decides

- Turn ON AC
- Close windows
- Reduce fan speed
- Continue monitoring
until goal is achieved.

Another example

Goal

Reduce electricity consumption.

The agent

- Switches OFF unused lights
 - Turns OFF TV after inactivity
 - Adjusts AC automatically
-

Advantages

- Intelligent planning

- Flexible
- Better decision making
- Goal oriented

Disadvantages

- Slower than reflex agents
- Requires planning algorithms

4. Utility-Based Agent**Definition**

A Utility-Based Agent selects the action that provides maximum overall benefit (utility).

Instead of just achieving goals,
it chooses the best possible solution.

Example

Suppose

Temperature = 29°C

Electricity price is high.

The user likes

24°C.

The utility-based agent decides

Instead of cooling to 22°C,

maintain

25°C

because

- Saves electricity
- Maintains comfort
- Reduces electricity bill

Another example

At night

The system

- Turns OFF unnecessary lights
 - Maintains minimum lighting
 - Activates security cameras
 - Optimizes battery backup
- This provides the highest utility.

Advantages

- Best overall decision
- Balances multiple objectives
- Learns user preferences
- Maximizes comfort and efficiency

Disadvantages

- Most complex
- Requires utility calculation
- Higher computational cost

Comparison of Intelligent Agents

Agent Type	Memory	Learning	Planning	Best Use
Simple Reflex	No	No	No	Light control
Model-Based	Yes	Limited	No	Security monitoring
Goal-Based	Yes	Yes	Yes	Temperature control
Utility-Based	Yes	Yes	Yes	Complete smart home automation

Hybrid Approach

A Hybrid Agent combines the strengths of different intelligent agents.

For example

Simple Reflex

↓

Turns ON lights immediately.

Model-Based

↓

Remembers user schedule.

Goal-Based

↓

Maintains desired temperature.

Utility-Based

↓

Optimizes electricity usage.

All four work together.

Why Hybrid Approach is Best

A hybrid system provides

- Fast response
- Better planning
- User preference learning
- Energy optimization
- Enhanced security
- Reduced electricity bills
- Improved comfort
- Higher reliability

Applications

- Smart lighting
 - Smart security
 - Smart thermostat
 - Voice assistants
 - Smart irrigation
 - Smart parking
 - Healthcare monitoring
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Advantages of Smart Home Automation

- Improved comfort
 - Better security
 - Saves electricity
 - Remote monitoring
 - Automatic scheduling
 - Learns user habits
 - Reduces manual work
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Disadvantages

- High installation cost
 - Internet dependency
 - Privacy concerns
 - Cybersecurity risks
 - Maintenance cost
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Conclusion

A Smart Home Automation System becomes highly efficient when multiple intelligent agents work together. While Simple Reflex Agents provide quick responses, Model-Based Agents maintain environmental knowledge, Goal-Based Agents achieve desired objectives, and Utility-Based Agents optimize comfort, energy efficiency, and security. Therefore, a **Hybrid Intelligent Agent** is the most effective solution because it combines speed, planning, learning, and optimization, making the home smarter, safer, and more energy-efficient.

Question 2

Robot Navigation in an Unknown Maze Using Uninformed Search Algorithms

Introduction

Robot navigation is one of the most important applications of Artificial Intelligence. In this problem, a robot is placed inside an unknown maze and must reach the exit without prior knowledge of the environment. Since the robot does not know the layout beforehand, it must explore the maze while making decisions based on available information.

Uninformed search algorithms do not use heuristic information. They search for solutions using only the problem definition. Two important uninformed search algorithms are **Uniform Cost Search (UCS)** and **Iterative Deepening Search (IDS)**.

Problem Description

The robot starts from a source location and must reach the exit while avoiding obstacles. Every movement has a cost, and the robot explores the maze until it finds the destination.

Possible movements:

- Up
- Down
- Left
- Right

Goal:

Find the exit with minimum cost.

Uniform Cost Search (UCS)

Definition

Uniform Cost Search expands the node with the **lowest cumulative path cost** first. It guarantees finding the least-cost path when all costs are non-negative.

Working Steps

1. Insert the start node into a priority queue with cost 0.
 2. Remove the node with the lowest path cost.
 3. Expand all valid neighboring nodes.
 4. Update cumulative costs and insert neighbors into the queue.
 5. Repeat until the goal node is reached.
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Example

S --1--> A --2--> G
 \--4--> B --1--> G

Possible paths:

- $S \rightarrow A \rightarrow G = 3$
- $S \rightarrow B \rightarrow G = 5$

UCS selects $S \rightarrow A \rightarrow G$ because it has the minimum total cost.

Advantages

- Guarantees optimal solution.
- Complete search algorithm.
- Works well when movement costs differ.

- Suitable for weighted graphs.

Disadvantages

- High memory consumption.
- Slower in large search spaces.
- Explores many nodes.

Time Complexity

Approximately $O(b^{(l + C/\epsilon)})$, where b is branching factor, C is optimal cost, and ϵ is minimum step cost.

Space Complexity

Similar exponential complexity due to storing frontier nodes.

Iterative Deepening Search (IDS)

Definition

IDS combines the space efficiency of Depth-First Search with the completeness of Breadth-First Search. It repeatedly performs depth-limited searches with increasing depth limits.

Working Steps

1. Set depth limit = 0.
2. Perform Depth-Limited Search.
3. If the goal is not found, increase the depth limit.
4. Repeat until the goal is reached.

Example

Depth 0:

S

Depth 1:

S

├── A
└── B

Depth 2:

S

├── A
│ ├── C
│ └── D
└── B
 ├── E
 └── G

The goal **G** is found at depth 2.

Advantages

- Requires very little memory.
- Complete for finite search spaces.
- Finds shallow solutions efficiently.

Disadvantages

- Repeats exploration of upper levels.

- Not optimal when path costs vary.
- Increased computation due to repeated searches.

Time Complexity
 $O(b^d)$

Space Complexity
 $O(b \times d)$

Comparison Between UCS and IDS

Feature	Uniform Cost Search	Iterative Deepening Search
Search Type	Cost-based	Depth-based
Data Structure	Priority Queue	Stack (Depth-Limited Search)
Optimal	Yes	Only with equal costs
Complete	Yes	Yes
Memory Usage	High	Low
Best For	Weighted graphs	Large unknown spaces

Intelligent Agent Design

Simple Reflex Agent

Responds only to the current location and cannot remember explored paths. Poor choice for unknown mazes.

Model-Based Agent

Maintains an internal map of explored locations and avoids revisiting known dead ends.

Goal-Based Agent

Plans movements toward the exit while considering current knowledge of the maze.

Utility-Based Agent

Evaluates multiple possible paths and selects the one that provides the highest benefit, considering travel cost, time, and safety.

Best Combination

The most effective solution is a **Goal-Based Agent combined with Uniform Cost Search (UCS)**.

Justification

- UCS guarantees the least-cost path.
 - The Goal-Based Agent continually plans toward the exit.
 - The internal model updates as new parts of the maze are discovered.
 - This combination minimizes travel cost while ensuring the robot reaches the destination efficiently and safely.
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Real-World Applications

- Warehouse robots
- Autonomous vehicles
- Search and rescue robots

- Planetary exploration
- Drone navigation
- Industrial automation

Conclusion

For navigation in an unknown maze, both UCS and IDS are valuable uninformed search algorithms. UCS is preferred when minimizing path cost is essential, while IDS is suitable when memory resources are limited. A **Goal-Based Intelligent Agent using Uniform Cost Search** offers the best balance of optimality, intelligent decision-making, and adaptability, making it highly suitable for real-world robotic navigation and autonomous systems.